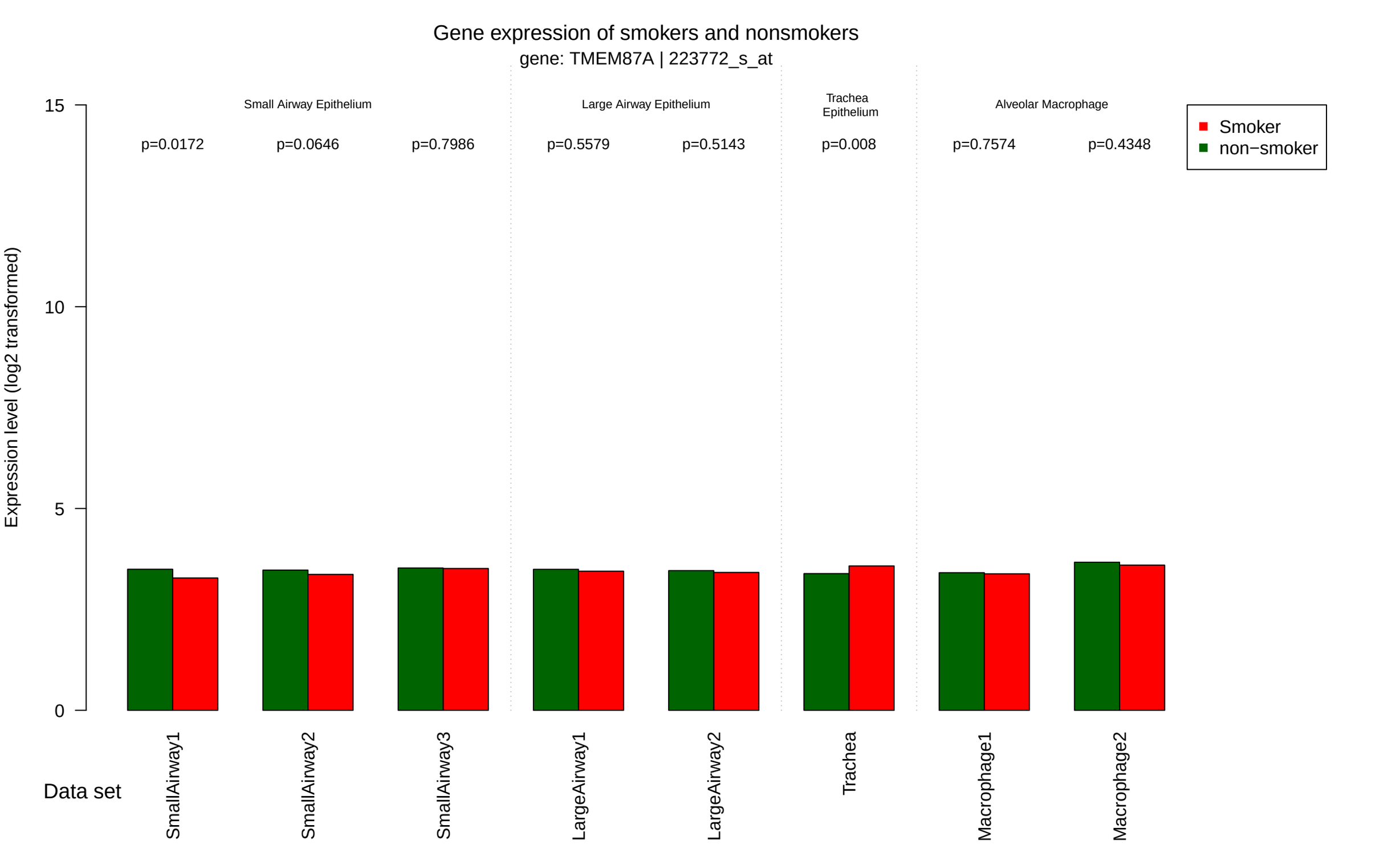
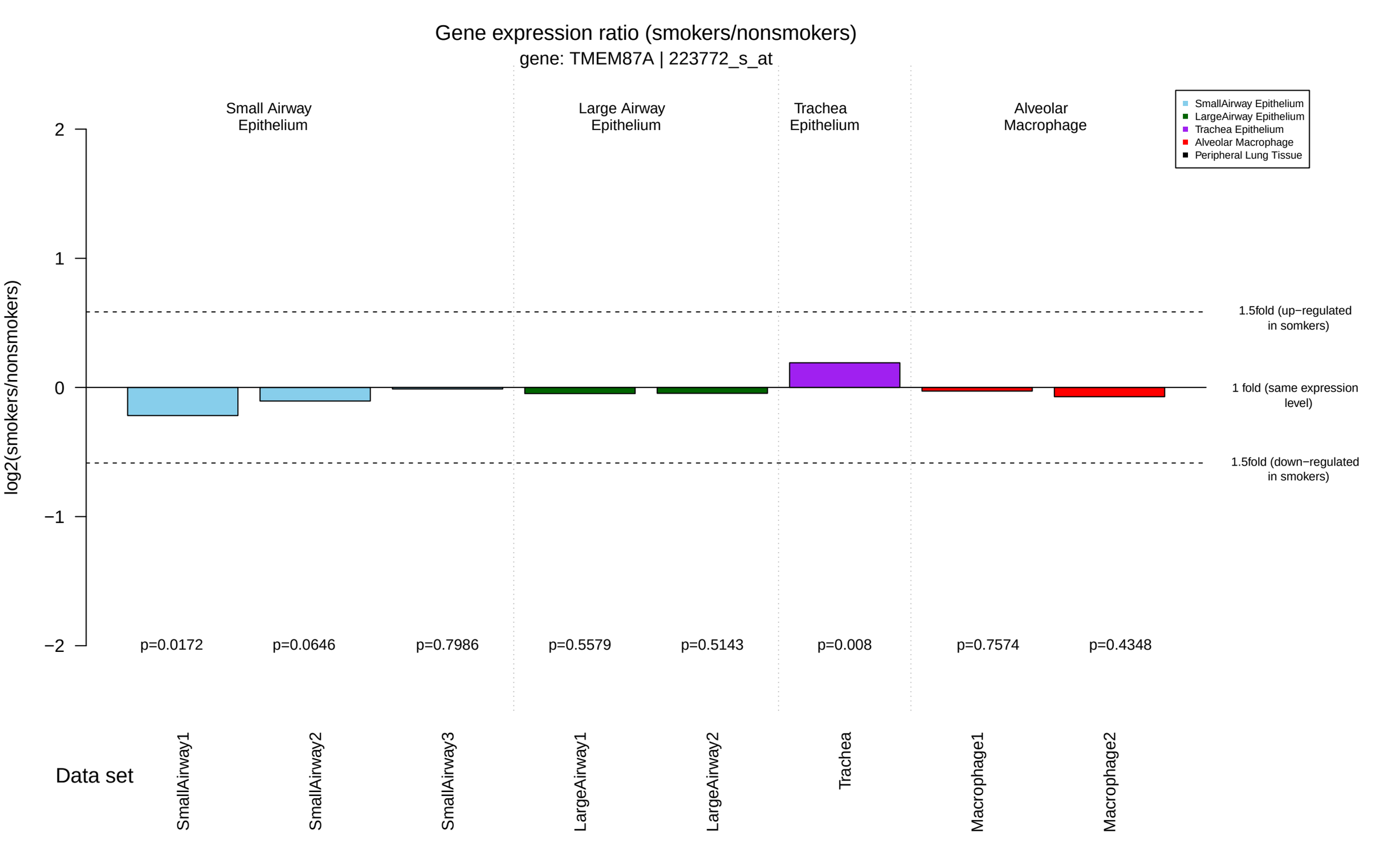
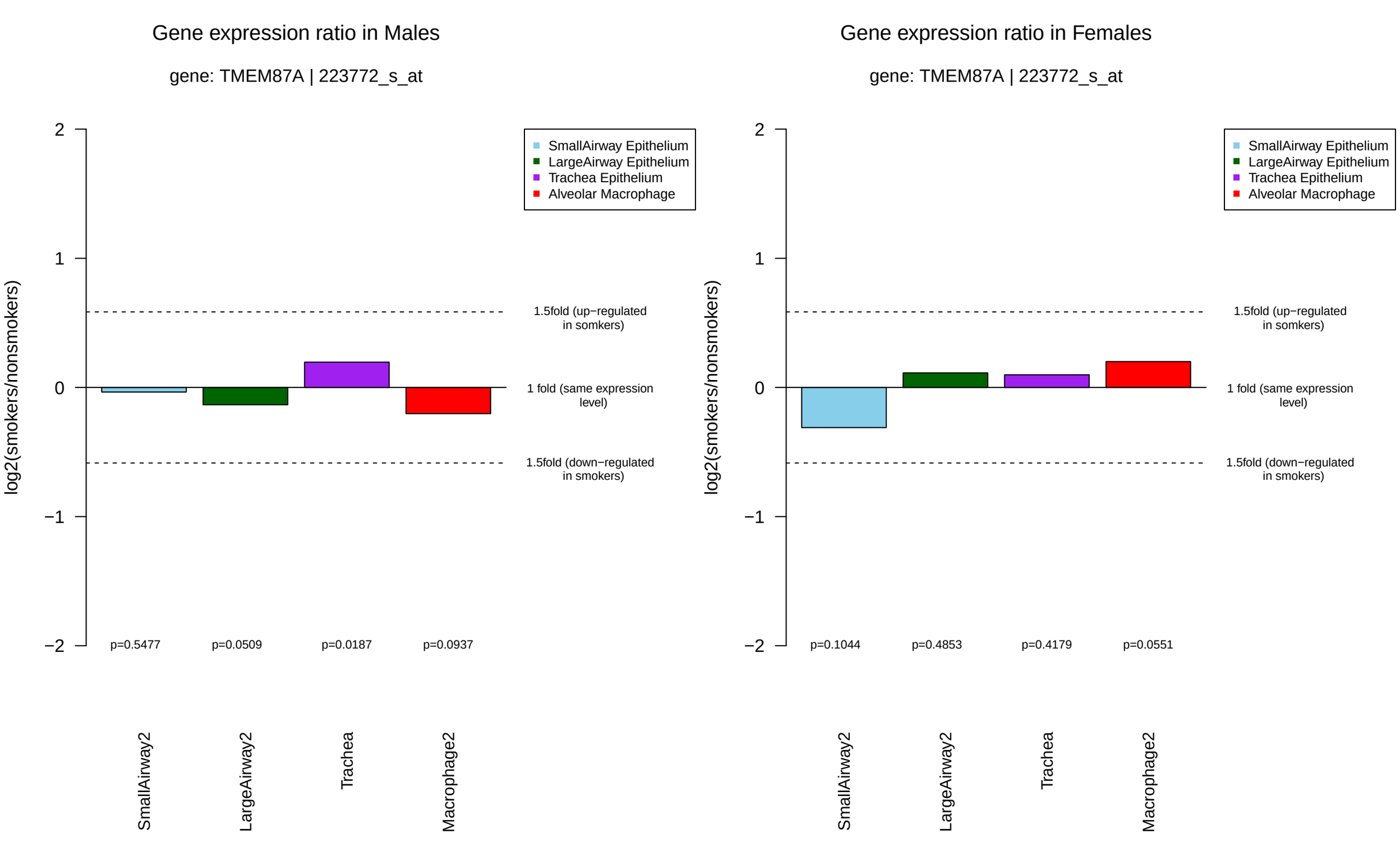
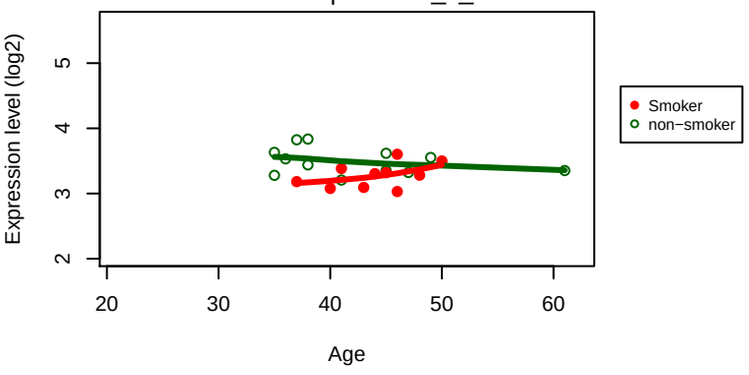
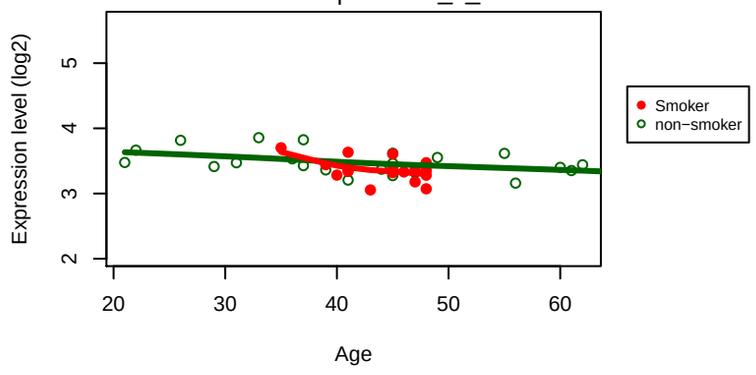
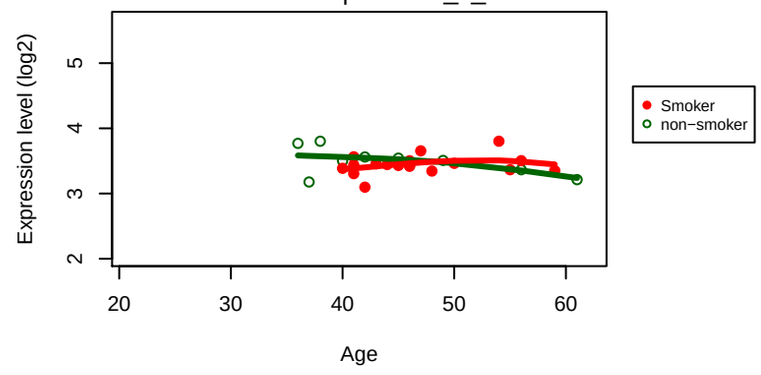

Gene expression of smokers and nonsmokers
gene: TMEM87A | 223772_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0172
p=0.0646
p=0.7986
p=0.5579
p=0.5143
p=0.008
p=0.7574
p=0.4348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

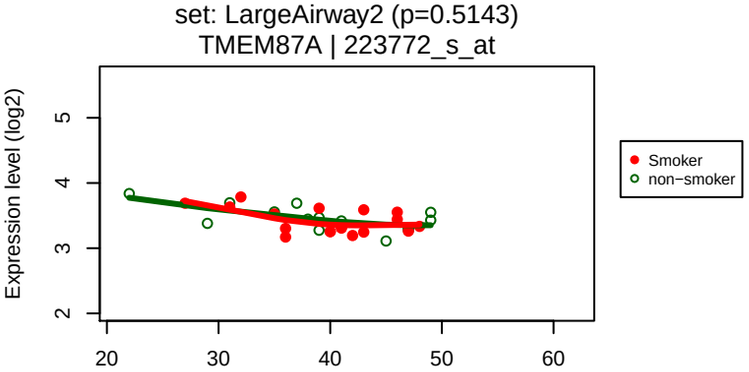

Gene expression ratio (smokers/nonsmokers)
gene: TMEM87A | 223772_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0172
p=0.0646
p=0.7986
p=0.5579
p=0.5143
p=0.008
p=0.7574
p=0.4348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

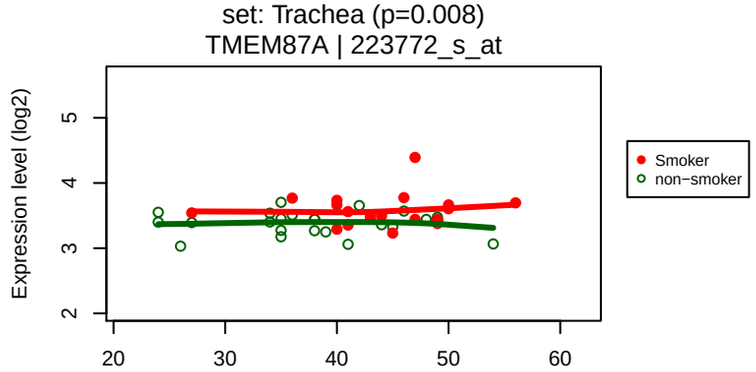

Gene expression ratio in Males
gene: TMEM87A | 223772_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5477
p=0.0509
p=0.0187
p=0.0937
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMEM87A | 223772_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1044
p=0.4853
p=0.4179
p=0.0551
SmallAirway2
LargeAirway2
Trachea
Macrophage2

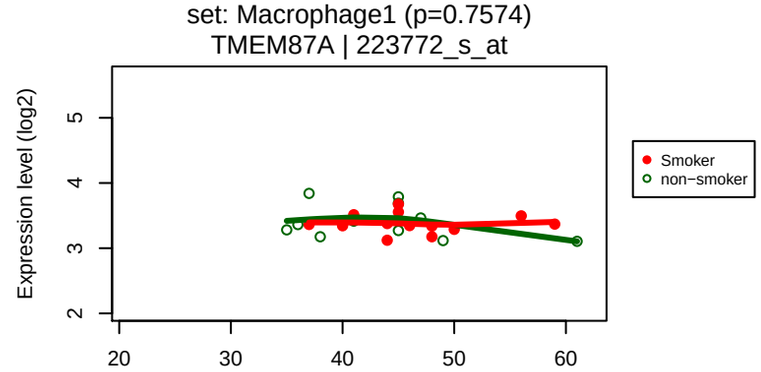

set: SmallAirway1 (p=0.0172)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker

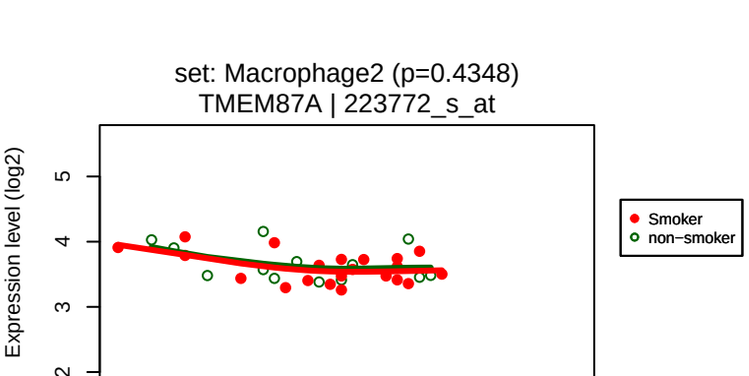

set: SmallAirway2 (p=0.0646)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker

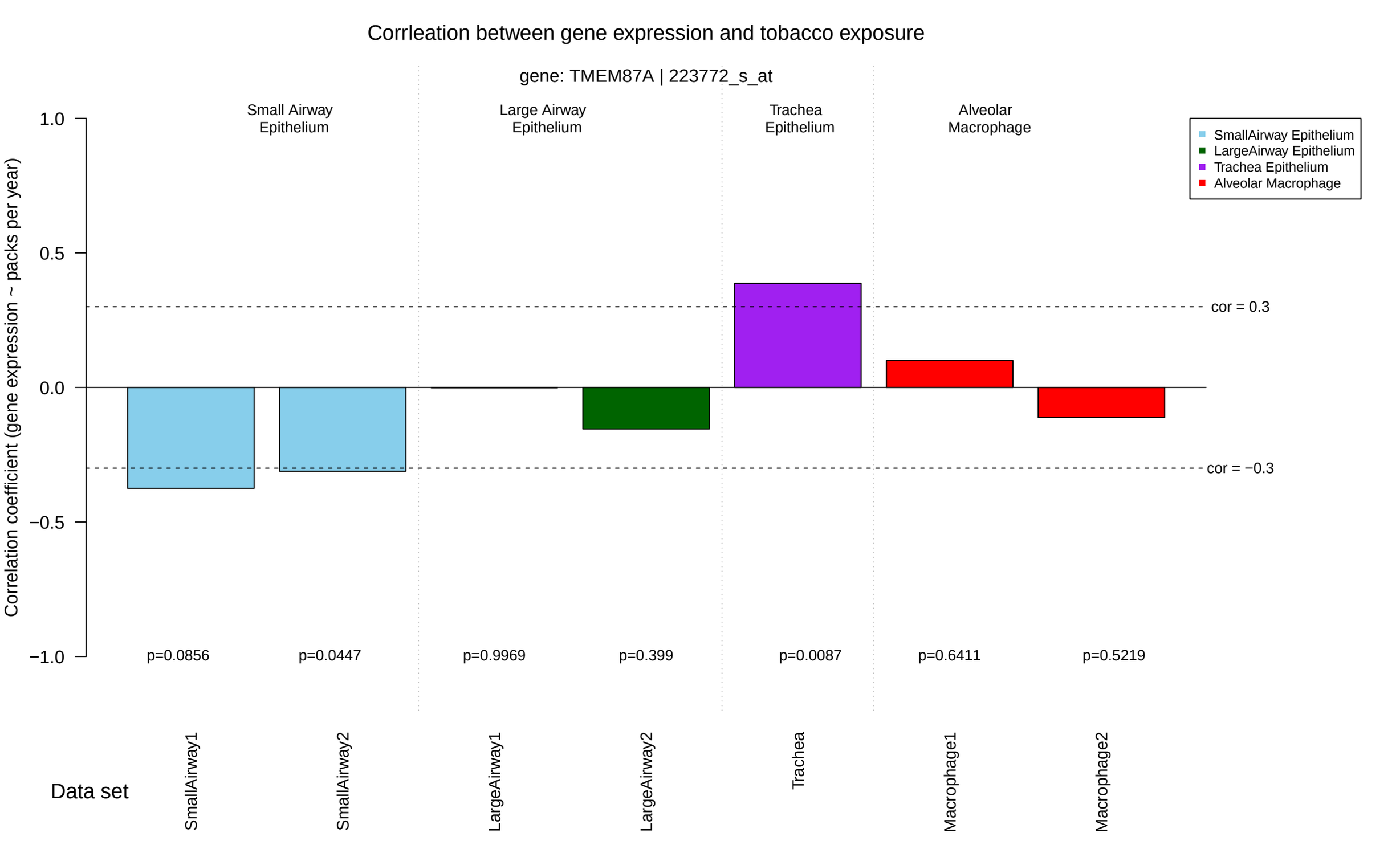

set: LargeAirway1 (p=0.5579)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5143)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.008)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7574)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4348)
TMEM87A | 223772_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMEM87A | 223772_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0856
p=0.0447
p=0.9969
p=0.399
p=0.0087
p=0.6411
p=0.5219
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2